

Inference

Last time

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

$$H_0 : \beta_2 = 0 \quad H_A : \beta_2 \neq 0$$

A new research question

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Research question: Is there a relationship between school rank and grad school acceptance, after accounting for GPA and GRE score?

What are my hypotheses for this new research question?

Testing multiple parameters

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

How do we test multiple parameters in a *linear* regression model?

Nested models

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

Full model (H_A):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

Reduced model (H_0):

Comparing nested models

Full model (H_A):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

Reduced model (H_0):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i$$

How should we compare these two nested models?

Comparing nested models

Deviance: for a logistic regression model, deviance is $-2 \log L$

Idea:

- + Maximizing likelihood (how we fit the model) is equivalent to minimizing deviance
- + Smaller deviance implies better fit
- + Is there a big difference in the deviance (model fit) between these models??

Change in deviance:

Comparing deviance

```
m_full <- glm(admit ~ gpa + gre + as.factor(rank),  
              family = binomial, data = grad_app)  
  
m_reduced <- glm(admit ~ gpa + gre,  
                 family = binomial, data = grad_app)  
  
m_reduced$deviance - m_full$deviance
```

```
## [1] 21.82649
```

How do I know whether this is a big change in deviance? That is: is a change in deviance of **21.83** unusual under the null hypothesis?

Parametric bootstrap

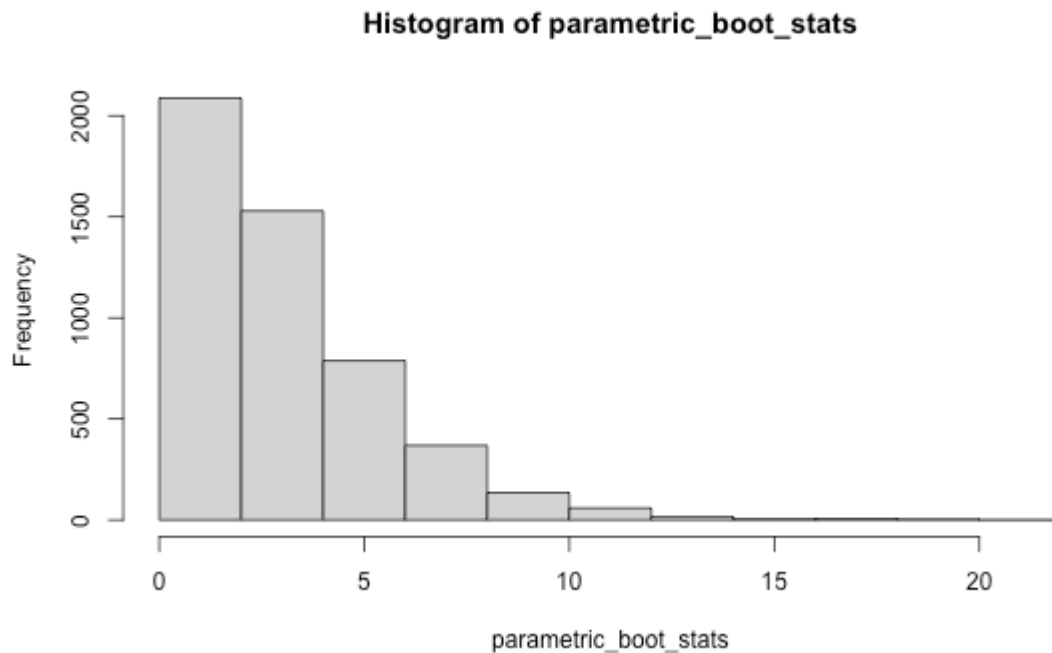
Class activity

- + Work on the class activity (linked on Canvas and course website)
- + Submit rendered HTML on Canvas at the end of class

Activity

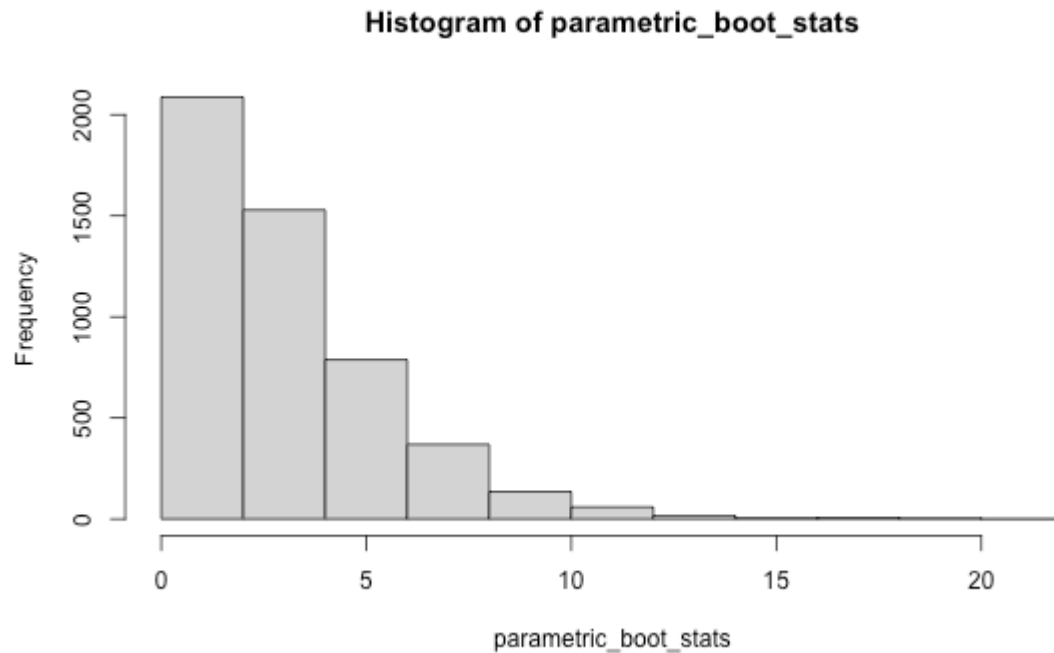
What does this distribution look like?

Activity



How unusual is the observed test statistic of 21.83, compared to this null distribution?

p-value



```
mean(parametric_boot_stats > 21.83)
```

```
## [1] 0
```